

The bones of humans and other vertebrates are alike in that they:

- ☐ A. consist solely of living tissue. [2%]
- ☐ B. develop from the embryonic ectoderm. [5%]
- ☐ C. form a rigid outer covering that protects the soft tissues inside the organism. [13%]
- ✔ ☒ D. form an internal support structure that serves as a scaffold for soft tissues. [78%]

Correct

78% answered correctly

Explanation:

Endoskeleton versus exoskeleton	
Endoskeleton	Exoskeleton
	
Consists of both living components (eg, osteocytes, osteoblasts, osteoclasts) and nonliving components (eg, calcium phosphate)	Consists of only nonliving components (eg, chitin)
Forms an internal scaffold that protects soft tissues (eg, organs) and serves as an attachment point for skeletal muscle	Forms a hard outer covering that surrounds and protects soft tissue underneath
Grows as the organism grows	Is shed and regenerated as the organism grows

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Vertebrates are a group of organisms that have an internal skeleton (endoskeleton) with a bony vertebral column that surrounds the spinal cord and provides points of attachments for other bones and muscles. The endoskeleton provides an *internal support structure* (scaffold) that allows for movement and protects the **internal organs**.

The vertebrate endoskeleton is composed primarily of **bone** but also contains **cartilage** in areas where flexible skeletal tissue is necessary. As a type of **connective tissue**, bone comprises organic and inorganic materials along with the various cell types (eg, osteoblasts, osteoclasts, osteocytes) involved in **bone remodeling**. In this scenario, the similarities between the bones of humans and other vertebrates must be compared. Accordingly, the bones of **all vertebrate** animals form an **endoskeleton** that provides a **scaffold for soft tissues** (eg, internal organs, skeletal muscle).

(Choice A) The bones of vertebrate organisms are composed of both *living* cells and *nonliving* materials (eg, collagen, calcium phosphate).

(Choice B) The embryonic ectoderm is one of three **germ layers** that develops into several structures, including the epidermis and nervous tissue. However, it is the embryonic mesoderm, *not* the ectoderm, that gives rise to the endoskeleton.

(Choice C) In contrast to vertebrates, invertebrate animals (eg, insects, crustaceans, worms) lack a vertebral column, but many possess an external skeleton known as an exoskeleton. This exoskeleton is a hard and rigid outer covering composed *solely* of nonliving material and serves to protect soft tissues underneath. Unlike the bones of vertebrates, exoskeletons cannot grow to accommodate organismal growth and must be shed and regenerated as the organism grows.

Educational objective:

The vertebrate endoskeleton is composed of both bone and cartilage and provides an internal scaffold that facilitates mobility while protecting and supporting internal organs. In contrast to vertebrates, many invertebrate animals possess an external skeleton (exoskeleton), which is a rigid outer covering that serves to protect the soft tissues underneath.

Time spent:0 sec

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Biology

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System:
Musculoskeletal System